# York University @ NeurIPS 2025 CURE-Bench Competition on Benchmarking AI Reasoning for Therapeutic Decision-Making

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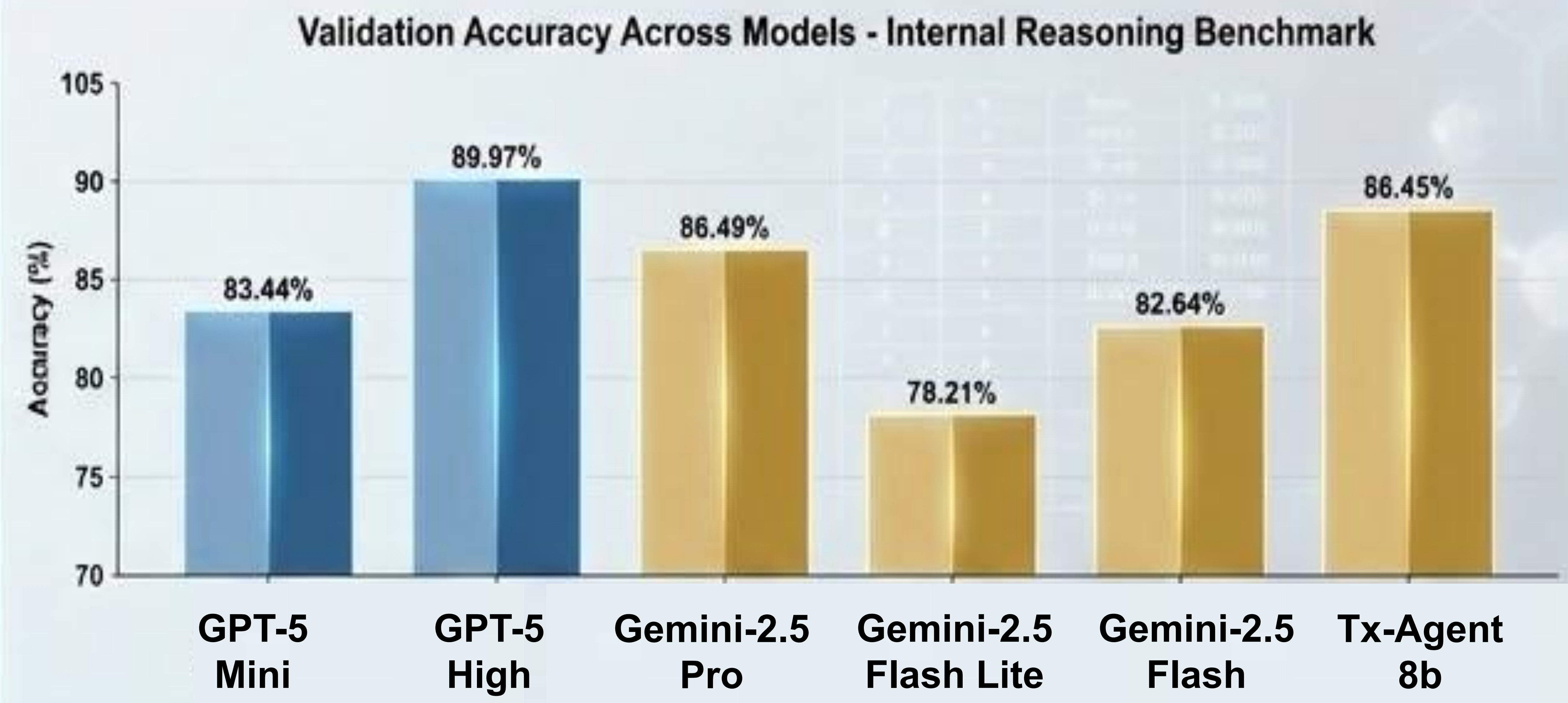
## Introduction

**Problem Statement:** Precision therapeutics require AI models that reason over complex patient-disease-drug relationships. Existing benchmarks fail to evaluate real-world clinical decisions.

**Competition Tracks:**

- **Internal Reasoning:** Model reasons using internal knowledge.
- **Agentic Reasoning:** Model Reasons via leveraging external tools.




## Model Selection Criteria

Evaluate multiple LLMs in the validation set of the internal reasoning benchmark to identify which models achieve the highest accuracy.

## Award Winning Model Results

**GPT-5 with High Reasoning Model:**

- Internal Reasoning: 69.38
- Agentic Reasoning: 74.43

## Methods

**Clinical Question**
MCQ / Open-ended

→ **GPT-5 High Reasoning**
Multi-step logic

→ **Web Search Agent**
40+ trusted domains

→ **JSON Output**
Reasoning + Answer